survivors, and assessing infrastructure damage, especially in regions inaccessible to manned aircraft.

3.4.3 Civilian and commercial applications

- *Aerial photography and mapping*: The inclusion of high-resolution cameras and GPS systems allowed UAVs to be used for topographical mapping, urban planning, and infrastructure documentation.
- *Agriculture and precision farming*: UAVs began to be adopted in agricultural monitoring, analyzing crop health, irrigation patterns, and pest infestations using multispectral imaging and environmental sensors.
- *Surveying and land inspection*: In civil engineering and land development, drones were utilized to inspect bridges, pipelines, and construction sites, reducing manual labor and safety risks.
- *Scientific research and exploration*: Drones provided cost-effective tools for geological surveys, oceanographic research, and remote sensing, supporting data collection in hazardous or hard-to-reach environments.

3.4.4 Communication and technological applications

- *Data transmission and relay operations*: Second-generation drones incorporated digital telemetry systems capable of transmitting real-time data between remote stations, making them effective in field research and mobile network expansion.
- *Technology demonstration platforms*: UAVs served as testbeds for developing autopilot systems, sensor integration, and flight control algorithms, which directly contributed to the design of third-generation and autonomous drones.